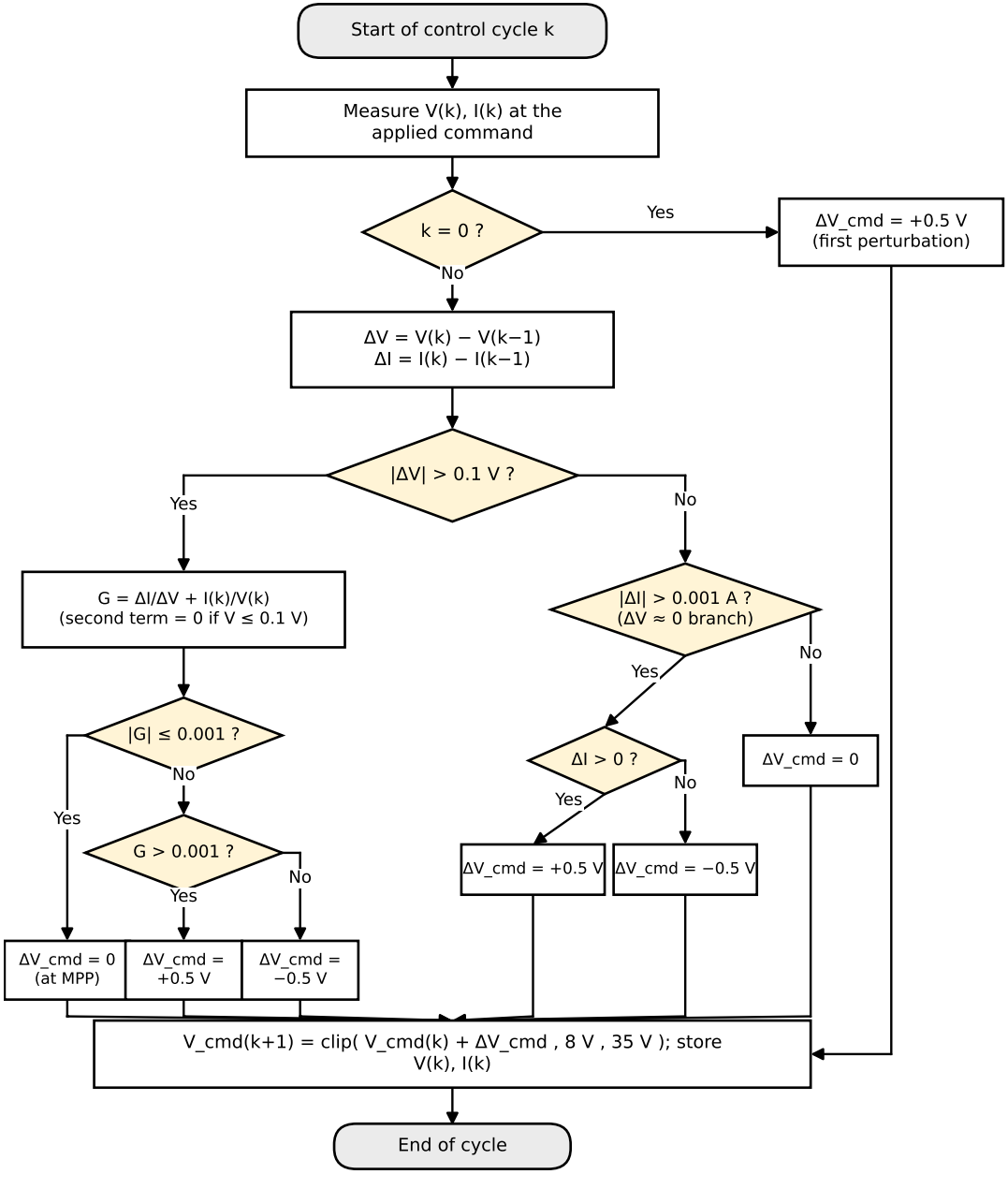
Start of control cycle k
Measure V(k), I(k) at the applied command
k = 0 ?
Yes
ΔV_cmd = +0.5 V (first perturbation)
No
ΔV = V(k) − V(k−1)
ΔI = I(k) − I(k−1)
|ΔV| > 0.1 V ?
Yes
No
G = ΔI/ΔV + I(k)/V(k) (second term = 0 if V ≤ 0.1 V)
|ΔI| > 0.001 A ? (ΔV ≈ 0 branch)
Yes
No
|G| ≤ 0.001 ?
No
ΔI > 0 ?
Yes
No
ΔV_cmd = 0
Yes
G > 0.001 ?
Yes
No
ΔV_cmd = +0.5 V
ΔV_cmd = −0.5 V
ΔV_cmd = 0 (at MPP)
ΔV_cmd = +0.5 V
ΔV_cmd = −0.5 V
V_cmd(k+1) = clip(V_cmd(k) + ΔV_cmd , 8 V , 35 V); store V(k), I(k)
End of cycle